

Secure and Robust Two-Phase Image Authentication

Seyed Amir Hossein Tabatabaei, Obaid Ur-Rehman, Natasa Zivic, and Christoph Ruland

Abstract—A novel two-phase robust content-based image authentication scheme is introduced. The proposed scheme is constructed based on a combination of hard and soft authentication using two existing generic approximate message authentication codes (AMACs). The AMACs combine error-correcting codes with cryptographic primitives such as message authentication codes and symmetric encryption algorithms. The message authentication codes are used for hard authentication, whereas the error-correcting codes introduce a certain degree of robustness in authentication. This is achieved by correcting minor unintentional modifications as a result of common image processing operations such as quantization, compression, and noise addition. The two-phase image authentication scheme verifies the authenticity of an image in two phases. The low frequency elements of the image in a transform domain are subjected to the first phase while some higher frequency elements are left to the second phase if the first phase succeeds. The proposed scheme tolerates common content-preserving modifications in an image but can discriminate intentional modifications affecting the image content. Mathematical bounds for the accuracy and the security level of the proposed approach are estimated and the performance is compared with some other well-known schemes in the literature. The results demonstrate that the proposed scheme shows high discriminating capability and can detect different types of meaningful forgery attacks on images while preserving the robustness. It also outperforms the benchmark image authentication schemes in terms of tradeoff between robustness and fragility.

Index Terms—Approximate message authentication code (AMAC), discrete cosine transform (DCT), discrete wavelet transform (DWT), error-correcting code, image authentication, unconditional security.

I. INTRODUCTION

MULTIMEDIA authentication has recently gained considerable attention due to the readily available multimedia editing and reconstruction tools and the need for transmission of multimedia objects, such as audio video and text, over non-secure networks. Authentication of these multimedia objects is very different from standard data authentication. Multimedia data is almost always preprocessed or modified before usage or transmission over (insecure) networks. The preprocessing

and modifications include operations such as compression, quantization, scaling, cropping etc. These operations affect the multimedia data; however, the content remains unaffected. Standard authentication mechanisms are effective in authenticating the actual multimedia data but not in authenticating the multimedia content. A new method for robust image authentication is discussed and developed in this paper. In general, the purpose of image authentication is to verify the authenticity of the images as long as the content is preserved. The level of content perseverance is defined by the content based authentication scheme in use, e.g., using a threshold value. This threshold is based on the criticality level of the application for which the threshold is chosen. Highly critical applications have more stringent requirements and therefore prefer a hard authentication by choosing a smaller threshold. Less critical applications use a soft authentication scheme by selecting a higher value for threshold. However, security should not be compromised by such mechanisms and the forged images or intentionally modified images with different content should be detectable. Image authentication is required to protect an image's copyright from gratuitous piracy [1]. Besides robustness and security which will be defined formally later, image authentication schemes may also satisfy certain auxiliary requirements. In the proposed scheme the authenticity of an image is verified in two phases with the aim of providing a sufficient level of robustness and security. The architecture is based on the combination of hard and soft authentication techniques provided by two existing individual authentication schemes constructed from cryptographic primitives and an error-correcting code. The error-correcting code is used for correcting unintentional modifications, and provides robustness against content-preserving changes. By adapting certain parameters in the proposed scheme, a different degree of robustness can be achieved based on the application requirements. The 2-phase image authentication scheme enables forgery detection on images with high probability. It can detect single and multiple meaningful forgery attacks on images outperforming comparable schemes significantly without reducing robustness. The latter is achieved due to a high level of protection afforded to the low frequency elements in the transform domain. It also provides an acceptable tradeoff between robustness and discriminating capability.

This paper is organized as follows. Section II introduces the required information, definitions and the problem formulation. The specification of the introduced image authentication scheme is discussed in Section III. The implementation details, simulation results and a comparison with other similar schemes are given in Section IV and Section V. Section VI contains the security and performance analysis. Section VII concludes the paper.

Manuscript received June 23, 2014; revised January 23, 2015 and April 26, 2015; accepted April 27, 2015. Date of publication May 13, 2015; date of current version June 13, 2015. The associate editor coordinating the review of this manuscript and approving it for publication was Prof. Jing-Ming Guo.

The authors are with the Chair for Data Communications Systems, University of Siegen, Siegen 57076, Germany (e-mail: amir.tabatabaei@uni-siegen.de; obaid.ur-rehman@uni-siegen.de; natasa.zivic@uni-siegen.de; christoph.ruland@uni-siegen.de).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TMM.2015.2432672

II. BACKGROUND

A. Image Authentication Methods

To verify image authenticity using standard image authentication techniques, an authentication tag is calculated by the sender and appended to the compressed image or embedded in it. When the receiver decompresses the image, this may potentially result in a modified image which is usually identical to, or very close to, the original one [2]. The image is declared authentic or inauthentic by comparing the reliably received or the extracted appended tag from the decompressed image with the recalculated or the re-extracted tag. Both sender and receiver use a shared secret key.

Different classifications exist for image authentication methods according to their construction and functionality. In the most common classification, image authentication approaches are divided into two major groups including strict or hard image authentication and soft image authentication [3]. The hard image authentication schemes do not tolerate any changes in the image content. The only accepted modifications are changes which preserve the pixel values, such as lossless compression or format conversion [3]. The main techniques used in this group are based on fragile watermarking and standard cryptography [4]–[6]. Fragile watermarking consists of generating a watermark and inserting it into the image such that it reflects any changes occurred in the image. The verification is successful if the pixels are left unchanged. The authentication tag in this method is generated mainly based on a checksum, predefined logo, discrete cosine transform (DCT) or color components [4], [7]–[9]. Image authentication schemes based on standard cryptography use common cryptographic primitives as their internal building blocks. They may provide partial forgery localization capabilities but no image restoration is normally possible. The main methods in this group consist of tiling the image into sub-blocks and calculating the MAC of the image row-wise, column-wise or block-wise [10], [11].

The second group concerns soft image authentication where the malicious content manipulations are rejected and the rest are accepted [3]. Semi-fragile watermarking and content-based signatures are the main approaches in this category. The general technique for semi-fragile watermarking is to insert the computed watermark into the image such that specific image processing operations such as compression and noise addition can be tolerated by the authentication scheme while malicious alterations are detected and localized. Additionally, the restoration capability is possible in some proposed schemes. Vector quantization and block similarities are some of important methods for watermark generation in semi-fragile watermarking schemes [12], [13]. The latest approach in soft image authentication techniques, namely content-based signature image authentication, which extracts the semantic content of an image to generate a digital signature, has attracted the most attention compared to other approaches. Content can be extracted from block intensity histograms [14], edges [15], statistical features of the image [16], DCT, discrete wavelet transform (DWT) or Fourier transform (FT) coefficients [17], [18] etc. Depending on the extracted features of the image,

content-based signature image authentication schemes are mostly robust against JPEG/JPEG2000 compression, some geometrical transformations, filtering and brightening. Image authentication methods are not limited to the aforementioned categories and hybrid schemes also exist. One constructional method is to apply abstract robust authentication models for content extraction which is strictly categorized into the above classification explicitly. Approximate message authentication codes (AMACs) are prominent examples of such models. They refer to the primitives performing approximate authentication as the input message comes from a fuzzy space e.g., multimedia or biometric data. Further information can be found in [2], [10], [19]–[25].

B. Definitions

The AMACs are suitable authentication mechanisms which tolerate acceptable modifications to the input message. To tolerate such acceptable modifications, *robustness* must be introduced and provided by the primitives. This causes reduction of the *security* and the accuracy level. Also the sensitivity of the AMAC against different images or malicious image modifications which indicates its discriminating capability may be endangered. This capability of an AMAC is referred to *fragility*. To formalize the aforementioned concepts, suppose that $\mathcal{I}$ denotes a set consisting of original images and their modified versions (including content-preserved versions and manipulated or forged versions) and $\mathcal{T}$ the tag set. A robust image hashing algorithm H is a one way function which generates a tag t of the image $I \in \mathcal{I}$ based on a secret key k via $t = H(k, I)$. The distance function d together with threshold values ϵ_1 and ϵ_2 are used to formalize the requirements. Robustness is defined by one or more parameters together with a threshold below which a tag generation algorithm tolerates content-preserving image modifications such as compression, noise addition, Gamma correction, etc. Let I indicate the original image and I' denote a content-preserved image corresponding to I . The robustness R can be defined as follows [26] where Pr stands for probability:

$$R(\epsilon_1) = Pr(d(H(I, k), H(I', k)) \leq \epsilon_1), \forall I, I' \in \mathcal{I}. \quad (1)$$

The fragility F refers to the degree of discriminating capability of an image hashing algorithm. It introduces a threshold value in which the hashing algorithm distinguishes between a maliciously modified image and the original one or its content-preserved versions. If I'' indicates a content-modified version of the image I , similar to (1) the fragility is defined as follows [26]:

$$F(\epsilon_2) = Pr(d(H(I, k), H(I'', k)) \leq \epsilon_2), \forall I, I'' \in \mathcal{I}. \quad (2)$$

Also the fragility is an indication of the sensitivity of the tag generation algorithm against the modifications of the image content. Increasing the fragility will decrease the robustness and consequently the performance of the mechanism. This tradeoff is analyzed empirically via receiver operating characteristic (ROC) curve which depicts true acceptance rate (TAR) versus false acceptance rate (FAR) parametrically.

Unlike fragility and robustness, the security requirement of an image authentication scheme is twofold. The first aspect is

the resistance of the image authentication scheme against a collision attack, which attempts to deceive the receiver using a different image whose tag is the same (or similar) to the reference tag. Secondly, it refers to the difficulty of recovering the shared secret key. In a key recovery attack, the attacker can deceive the receiver by generating an image-tag pair of her own. More information is found in [26]–[28].

C. Two Generic AMAC Schemes

The most well known generic models for AMAC have been introduced in [2], [19]–[21]. Here, two approaches are used for the introduced image authentication scheme. In the first scheme [19], called AMAC₁, the authentication tag is created based on a systematic error-correcting code (*Enc*, *Dec*), a standard MAC algorithm $MAC(k_1, \cdot)$ and a symmetric encryption-decryption algorithm (E_{k_2} , D_{k_2}). $Enc(\cdot, \delta)(Dec(\cdot, \delta))$ is an encoding (decoding) algorithm with an error correction capability of δ symbols and E_{k_2} (D_{k_2}) indicates an encryption (decryption) algorithm. The verification part of AMAC₁ is the same as a hard authentication method but with the extra capability of correcting up to δ bit errors before verification [19]. The authentication tag in the second scheme, AMAC₂, is created based on the construction of probabilistic universal one-way hash functions [29]. The tag generation algorithm uses a pseudo-random function (PRF) $f_k(\cdot)$ and a target collision resistant hash function (TCR) $tcru(\cdot)$ as its building blocks. The verification algorithm of AMAC₂ performs soft authentication by evaluating the equality between the received and the computed hash-block sequences in a preset number of positions defined by the threshold Tr . The descriptions of the tag generation algorithm of AMAC₁ and AMAC₂ are given in Algorithms 1 and 2 below [19].

Algorithm 1 AMAC₁ tag generation

Input: shared secret keys k_1, k_2 generated by $Kg(\cdot, d)$, m -bit message x , error correction capability δ

```

1: procedure AMAC1 – Tag( $k_1, k_2, x$ )
2:    $x||pc \leftarrow Enc(x, \delta) \triangleright pc$  indicates the parity check
     value of the message.
3:    $T_1 \leftarrow MAC(k_1, x)$ 
4:    $C \leftarrow E_{k_2}(pc)$ 
5:    $tag \leftarrow C||T_1$ 
6:   return tag
7: end procedure
```

Algorithm 2 AMAC₂ tag generation

Input: shared secret key k generated by $Kg(\cdot, d)$, initial value IV , m -bit message x , block size c , number of blocks t , approximate correctness parameters P and δ , approximate security parameter γ

```

1:  $t_1 = \lceil m/2\delta \rceil, t_2 = \lceil -10 \log(1 - P) \rceil \triangleright$ 
   The messages having difference up to  $\delta$  bits are assumed
   identical by the user.
2: procedure AMAC2 – Tag( $k, x$ )
3:    $(u||\pi||\rho||L) \leftarrow PRF_k(IV) \triangleright u \in \{0, 1\}^k, \pi$  is a
     permutation function and  $L \in \{0, 1\}^m$ .
4:    $x_1||\dots||x_t \leftarrow \pi(L \oplus x)$ 
```

```

5:   for  $i = 1, t_2$  do
6:      $N_i \leftarrow x_{i_1}||\dots||x_{i_{t_1}} \triangleright \rho$  is used to select  $i_1, \dots, i_{t_1}$ 
       randomly from  $\{1, \dots, t\}$ .
7:      $T_i \leftarrow tcru(N_i) \triangleright tcru$  denotes a target collision
       resistant hash function.
8:   end for
9:    $tag \leftarrow IV||T_1||\dots||T_{t_2}$ 
10:  return tag
11: end procedure
```

Algorithm 1 performs hard authentication so, the typical value for ϵ_1 and ϵ_2 is zero. According to Algorithm 2, the expected robustness of the latter scheme is $R(t_2 Tr) = P$ while the corresponding fragility, $F(t_2 Tr)$ is roughly estimated by P, γ and δ [19]. Detailed description including security and performance analysis of the aforementioned schemes is given in [19].

III. PROPOSED 2-PHASE IMAGE AUTHENTICATION

In this section, a generic 2-phase scheme for content-based image authentication based on the initial work presented in [25] is introduced. The initially presented scheme is developed and significantly enhanced and, comprehensive experimental and comparison results are given. The above mentioned AMACs have advantages and drawbacks to be used individually in image authentication applications. The advantages and disadvantages of the latter schemes can be shortlisted as follows.

- AMAC₁
 - 1) (+) AMAC₁ enables error correction using a systematic error-correcting code.
 - 2) (+) Computational cost of AMAC₁ is low: AMAC₁ generates the approximate tag of the message by just one encoding, one encryption and one MAC operation.
 - 3) (+) Security analysis of AMAC₁ from the computational security viewpoint is given and different attack scenarios are considered [19]. This issue also applies to AMAC₂.
 - 4) (-) The length of AMAC₁ depends on the parameters of systematic error-correcting code and the desired correction capability which can result in generating a large tag size for long messages.
 - 5) (-) AMAC₁ does not perform soft authentication in terms of tolerating errors which are not corrected by the error-correcting code. This issue limits the robustness when a higher robustness level is desired.
 - 6) (-) Error localization is not possible in AMAC₁.
- AMAC₂
 - 1) (+) AMAC₂ performs soft authentication based on a predefined threshold value, and the level of desired accuracy and security can be adjusted. This represents a significant advantage of AMAC₂, where a lower bound for the threshold value can be set analytically.
 - 2) (+) The length of AMAC₂ is fixed with the same adjusting parameters for messages with different lengths.
 - 3) (+) AMAC₂ enables error detection and partial error localization.
 - 4) (-) AMAC₂ does not provide error correction capability.

Due to above advantages and drawbacks of the discussed AMACs, a hybrid design for an image authentication scheme is proposed here which inherits individual benefits of each scheme without exhibiting their drawbacks. The introduced scheme is based on two generic AMACs presented in [19] for performing hard and soft authentication task respectively. Protecting the parts of the extracted feature which are more fragile to the content modification by hard authenticating AMAC enables forgery detection, while supporting the less fragile content-holder parts provides robustness of the mechanism. The final authentication is successful when both of the verification algorithms corresponding to the two phases succeed.

A. Hash Generation Algorithm

An image hashing algorithm consists of preprocessing of image, intermediate hash generation containing feature extraction and final hash.

1) *Preprocessing*: The input image I is firstly normalized using bilinear interpolation and mapped to a fixed-size $m \times m$ image. To increase robustness against minor content-preserving modifications, an optional low-pass filtering such as a low-pass Gaussian filter can be applied as well [30]. The preprocessed image is called J ($J \in \mathbb{R}^{m \times m}$). Then the image J is divided into p^2 non-overlapping equal size blocks $J_1, \dots, J_{p^2} \in \mathbb{R}^{l \times l}$. Here $l = 8$ which is a good match with the JPEG compression, the DCT mapping and the image size 128×128 or 256×256 in use. Other values can be assigned for l in case of different image sizes or compression methods. A proper choice of l decreases the computational load of the tag generation algorithm.

2) *Intermediate Hash Generation*: The generic method for calculating the intermediate hash consists of first applying a transformation T on the image blocks $J_1, \dots, J_{p^2}$ to get the feature vector. $v_i = T(J_i)$ is computed for each image block and the elements containing the more compact information about the image such as low frequency components are extracted and denoted by $u_1, \dots, u_{p^2}$ to be supported by hard authentication mechanism AMAC₁. The elements are quantized to form a q -bit message. Let the tag generated in this step be denoted as tag_1 . tag_1 is generated from image feature parts which are tolerant and robust against minor content-preserving image modification. However when malicious manipulations or attacks on images occur the selected feature elements are influenced such that the hard authentication will fail with high probability. More robustness in this step can be achieved through increasing the error-correction capability of AMAC₁. A selection of other elements from v_i s containing less compact information about image are considered for the soft authentication step. The elements are used by AMAC₂ tag generation algorithm to generate the second tag tag_2 . The generated tag in this step is computed based on the probabilistic target collision resistant hash functions (Algorithm 2) which enables approximate or soft authentication when content-preserving modifications on image occur. The choice of transformation is specified according to the desired tolerated

modification and computational cost of the algorithm. For example the DCT or the DWT is used when the robustness against JPEG or JPEG2000 is the main concern [4], [25], [28]. Matrix transformations such as singular value decomposition (SVD) and nonnegative matrix factorization (NNMF) are considered when robustness against geometrical minor modification is desired. A combination of two transforms can also be used, as in the method proposed in [31]. If $T = DCT$ is used [25], then the DC coefficients are selected for AMAC₁ and ten AC coefficients are used by AMAC₂. In the case of $T = SVD$, the first right and left singular vectors are picked by AMAC₁ and the second left and right singular vectors are used by AMAC₂. A similar strategy can be used when other feature vectors are extracted from the image.

3) *Final Hash Vector*: Concatenating tag_1 and tag_2 forms the final hash vector tag of the image or the authentication tag submitted over a reliable channel to the receiver.

The formal description of the tag generation algorithm is given in Algorithm 3.

Algorithm 3 2-phase tag generation (proposed scheme)

Input: image I , image dimensions (in term of image blocks): M, N , input parameters of Algorithm 1, input parameters of Algorithm 2

- 1: Divide I into non-overlapping square blocks $b_{1,1}, \dots, b_{M,N}$ after image preprocessing operations.
 - 2: **procedure** 2 – phase – Tag(k_1, k_2, k_3, I)
 - 3: **for** $i = 1, M$ **do**
 - 4: **for** $j = 1, N$ **do**
 - 5: $A = T(b_{i,j})$
 - 6: $C_{i,j}^1 \leftarrow$ selected quantized elements of A for the first authentication phase
 - 7: $C_{i,j}^2 \leftarrow$ selected quantized elements of A for the second authentication phase
 - 8: **end for**
 - 9: **end for**
 - 10: $tag_1 = \text{AMAC}_1 - \text{Tag}(k_1, k_2, C_{1,1}^1 \parallel \dots \parallel C_{M,N}^1) \triangleright tag_1$ is the tag which is used for verification by the hard authentication scheme.
 - 11: $tag_2 = \text{AMAC}_2 - \text{Tag}(k_3, C_{1,1}^2 \parallel \dots \parallel C_{M,N}^2) \triangleright tag_2$ is the tag which is used for verification by the soft authentication phase.
 - 12: $tag = tag_1 \parallel tag_2 \triangleright tag$ is the generated final image authentication vector.
 - 13: **return** tag
 - 14: **end procedure**
-

B. Verification Algorithm

Let I' and tag denote the received image message and the received tag respectively. tag is decomposed into its components tag_1 and tag_2 corresponding to the received tags generated by AMAC₁ and AMAC₂. The verification process of the proposed scheme takes place in two phases. I' is preprocessed and divided

into image blocks $J'_1, \dots, J'_{p^2}$ (here $M = N = p$) and after applying T to extract $u'_1, \dots, u'_{p^2}$ and quantization, the MAC is computed using the secret key. If the calculated MAC matches the received one extracted from tag_1 , the first phase of the verification corresponding to $AMAC_1$ succeeds. Otherwise an attempt is made to correct $u'_1, \dots, u'_{p^2}$ using the error-correcting code. If the first verification phase succeeds, the second verification phase corresponding to $AMAC_2$ starts. The verification in the second phase is successful if at least $Tr.t_2$ positions in tag_2 have equal hash blocks. The choice of the threshold factor Tr will be discussed in Section VI-A. A rough lower bound is estimated for Tr such that the desired robustness and fragility hold.

IV. IMPLEMENTATION

Various implementations can be achieved via different selection of the transformation T . Here, the special case of $T = DCT$ is presented and implemented. A Reed Solomon (RS) code [32] is selected as the error-correcting code in the first phase. However other error-correcting codes can be used if they provide the desired error correction capability. The mechanism is called the DCT-RS-based scheme. Accordingly, when $T = DWT$ or $T = SVD$ it is called the DWT-RS-based and the SVD-RS-based scheme respectively.

Let $T = DCT$, the image I is firstly normalized using a bilinear interpolation and mapped into a fixed square size image. Then, a low-pass Gaussian filter is applied to create the preprocessed image J . J is divided into non-overlapping blocks and the 2-dimensional DCT is applied to each image block. The DC element from each block is picked and quantized. The quantized DCs are concatenated together to form a q -bit message for hard authentication part corresponding to $AMAC_1$. The choice of quantization coefficients for DCs depends on the expected robustness. Larger quantization coefficients provide more robustness against acceptable image modifications. A Reed Solomon code is used in the correction step. RS codes are defined over a Galois Field ($GF(p_1^{p_2})$) where p_1 is a small prime number with typical value 2 while p_2 is a small integer and typically $2 \leq p_2 \leq 8$. They have the ability to detect and correct multiple symbol errors.

When there are p^2 image blocks, DCs can be protected row-wise. Thus p codewords are generated, each of n_1 symbols. For a grey-scale image, the DC elements can be represented in 8-bits (after quantization), and therefore $RS(255, 223, 33)$ code is chosen for encoding in the $AMAC_1$ tag generation algorithm. As the chosen RS code has the ability to correct 16 symbol-errors, code shortening must be used, i.e., by appending 207 "0" symbols before RS encoding and not transmitting the "0" symbols as they can also be reinserted at the receiver without transmission. The second phase of the hash generation tag takes the first ten AC coefficients of the DCT matrix of each block according to zigzag order and quantization table used in the JPEG compression algorithm. The advanced encryption standard (AES) and secure hashing algorithm based MAC (HMAC-SHA-256) are selected for the encryption-decryption algorithm and the MAC respectively.

V. EXPERIMENTAL RESULTS

A. Robustness

To demonstrate the robustness of the proposed scheme against content-preserving image modifications, the input image is resized (if necessary) to 128×128 pixels and subjected to preprocessing operations by applying bilinear interpolation. An image database consisting of 110 images is selected from the self captured images and images downloaded from image databases available on the Internet. Each image undergoes a set of conventional content-preserving modifications. This set is categorized based on filtering operations, additive noise, JPEG compression and geometric distortions [33]. The filtering operations include median filtering ordered from 4 to 13, average filtering ordered between 1 and 8 and blurring ordered between 0.2 and 2. For the additive noise, Gaussian-distributed noise is considered with variance from 0.01 to 0.1, as well as speckle noise with variance lying between 0.05 and 0.5. The geometric distortions considered include scaling and shearing with corresponding factors ranging from 0.1 to 1, and 1 to 10 percentage respectively. Gamma correction applies in the range $\gamma \in [1.005, 1.05]$. The JPEG compression considered has a quality factor ranging from 20 to 80. For each image, 89 similarly distorted versions are generated according to the aforementioned modifications, resulting in 9790 images in total. The normalized Hamming distance between the hash of the original image and the distorted ones is calculated and the ROC curve is plotted based on the different threshold values. The false acceptance rate for all kinds of modifications is approximately zero, while the true acceptance rate varies according to the type of distortion. To calculate a true acceptance rate for a total of n_1 images (including original and content-preserved versions), let the number of content-preserved versions of the original images declared as authentic be n_2 . The true acceptance rate is calculated as n_2/n_1 . To calculate a false acceptance rate for a total of n'_1 images (including the original and the modified/attacked versions), the number of images judged as authentic images is calculated and denoted by n'_2 . The false acceptance rate is calculated as n'_2/n'_1 .

According to Fig. 1, the introduced scheme shows high robustness against JPEG compression, additive Gaussian noise, blurring and uniform filtering. The scheme shows lower robustness when the image is subjected to average filtering, gamma correction and shearing. However, the presented scheme does not tolerate rotation and scaling.

B. Towards Greater Robustness

It is known that increasing robustness by relaxing the existing parameters can endanger security. In order to establish robustness against rotation and scaling, a method based on the idea of ring partitioning introduced in [34] will be used. In this regard, after preprocessing, the image is divided into n rings $\mu_1, \dots, \mu_n$ by using concentric circles. Each ring contains an equal number of pixels. When some pixels are located in the circular border line between the rings, the number of pixels is made equal by using linear interpolation. The pixel values are then sorted to assure invariance against rotation and finally the transformation T is applied. A detailed description is given in [34]. Although

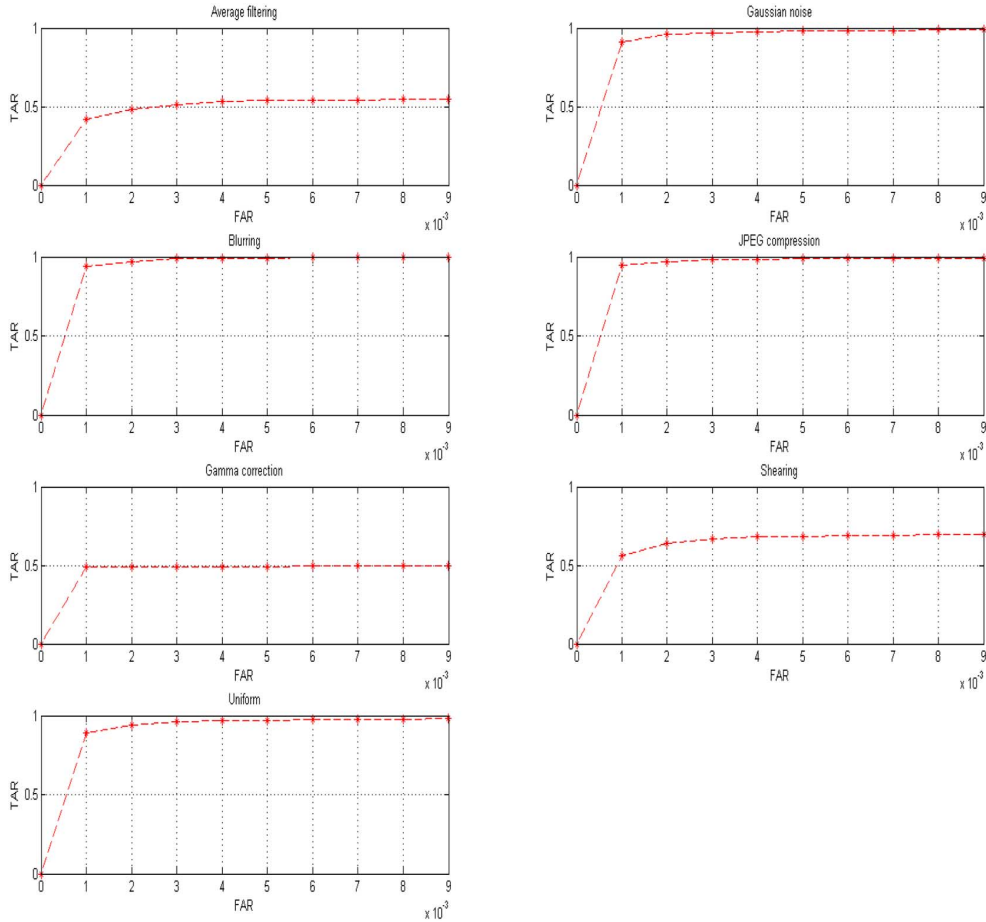


Fig. 1. ROC curves corresponding to the DCT-RS-based implementation for content-preserving image modifications.

this novel idea makes the image invariant against arbitrary rotation, the pixels in the area between the image boundaries and the inscribed circle are not covered. This results in ignoring approximately 21% of the image pixels by generating the intermediate hash vector. To overcome this deficiency, an out-scribed circle can be used. Thus after ring partitioning, an out-scribed circle is used for the whole image together with a padding scheme to fill the area between the circle and the original boundary of the image in order to assign an equal number of pixels to the other rings. The added values are set to a constant value (e.g., $\min(\text{pixel values}) - 1$) to be distinguished from the actual pixel values.

C. Fragility

Fragility of an image authentication scheme is an indication of discrimination capability. To evaluate the fragility of the proposed approach, an image set consisting of 1000 perceptually different images has been considered. The tag corresponding to each image has been computed and compared pair-wise with the other tags, which results in 499,500 comparisons. The threshold values are set according to the expected fragility extracted from adjusting the parameters of AMAC₂. The results are shown in Fig. 1. According to these results, when the threshold ratio Tr varies from 10% to 90%, the false acceptance varies from 0.0001 to 0.05, resulting in a good discrimination capability.

D. Comparison Results

Comparing the results to all existing image authentication mechanisms is not feasible here. Therefore, five schemes from different categories are selected, which include the main existing methods for image hash generation.

1) *DWT-Based Image Hashing*: This approach has been introduced by Venkatesan *et al.*, in [35]. To extract the feature of an image, at first a 2D-DWT is applied to the image and then each subband is randomly tiled using a secret key. The predefined statistics of each tile are calculated and rounded to produce a binary sequence. The final hash value is generated after applying the decoding stage of an error-correcting code to the binary sequence. The normalized Euclidean distance is used to evaluate the difference between generated tags. The optimal threshold value $Tr = 0.08$ is set according to Image Hashing Research project.¹

2) *SVD-Based Image Hashing*: Kozat *et al.* presented an SVD-based image hashing scheme in [31]. The image is randomly divided into possibly overlapping rectangles and the SVD of each rectangle is calculated. The first left and right eigenvector of each rectangle are used as the elementary feature of the images to form an intermediate vector. The intermediate vector is randomly mixed and tiled and then subjected to the

¹[Online]. Available: <http://users.ece.utexas.edu/~bevans/projects/hashing/index.html>

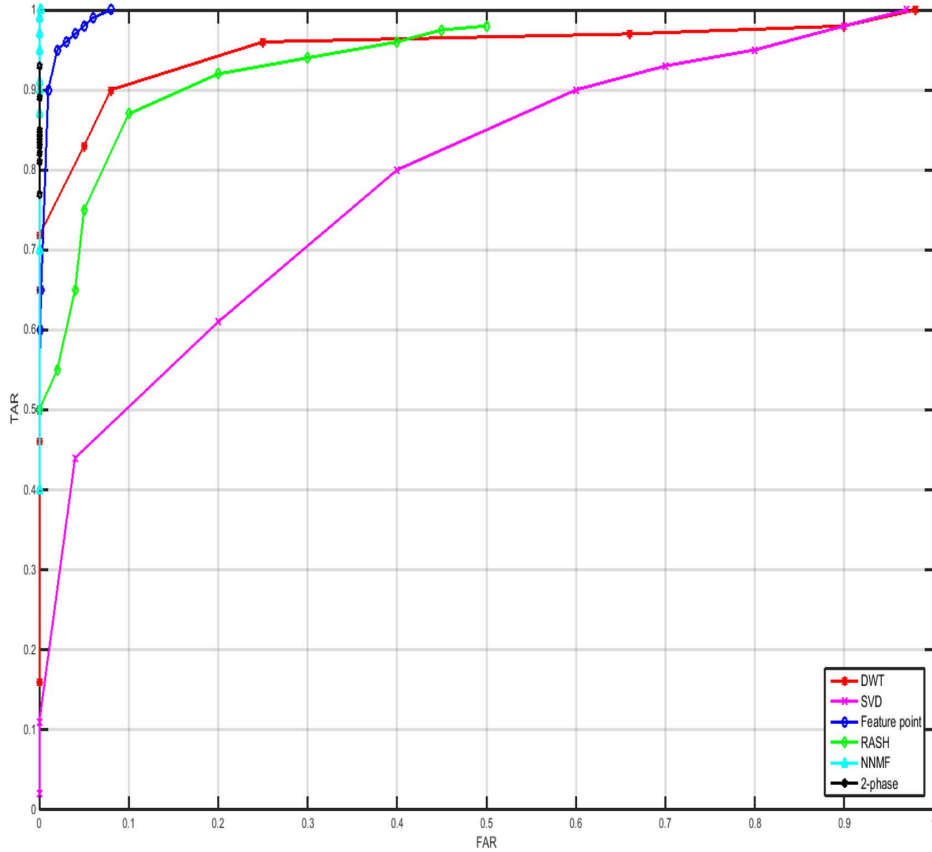


Fig. 2. ROC curve comparison for different schemes: DWT-based (“DWT”) image hashing of [35], SVD-based (“SVD”) image hashing of [31], feature point-based (“Feature point”) image hashing of [36], Radon-based (“RASH”) image hashing of [38], NNMF-based (“NNMF”) image hashing of [34] and the two-phase (“2-phase”) image authentication scheme. The tradeoff between robustness and fragility is depicted and compared in terms of various content-preserving image modifications including filtering operation, additive noise, compression, and geometric distortions applied on a sample of different images based on ten threshold values.

second SVD transformation. The final hash vector is generated by picking the first left and right eigenvector and concatenating them. Like the latter approach, the normalized Euclidean distance is used as a measure function with the optimal threshold value $Tr = 0.02$.

3) *Feature Point-Based Image Hashing*: Monga and Evans introduced an image hashing approach based on the feature points of an image in [36]. The tag generation algorithm produces the hash vector either in a deterministic way or in a probabilistic way. The deterministic algorithm is an iterative algorithm which looks onto the invariant feature points based on the existing algorithm [37]. The randomized algorithm uses a secret key to generate random locations in the image and then extracts the feature points corresponding to the randomly selected region. Combining the feature points gives the final hash vector. The normalized Hamming distance is used for the evaluation of the difference with the optimal threshold value $Tr = 0.2$ [36].

4) *Radon-Based Image Hashing*: Radon soft hash algorithm (RASH) has been proposed in [38] based on Radon transformation. The image is firstly subjected to Radon transformation in all directions (angles) and then the medium points are evaluated as feature points to generate the final hash vector. The peak value of the cross-correlation is used as a metric to evaluate the

similarity between hashes with the optimal value $1 - Tr = 0.87$ [30], [38].

5) *NNMF-Based Image Hashing*: The recently proposed image hash algorithm of [34] computes the hash vector based on non-negative matrix factorization. The tag generation algorithm firstly uses the aforementioned ring partitioning procedure on the image to produce the secondary image. Then the NNMF is applied and the coefficient matrix introduces the final hash vector. The correlation coefficient is used to evaluate the similarity between hashes. The authors of [34] recommended $1 - Tr = 0.95$ (or 0.98) for an optimal performance.

For a fair comparison, the same image set is used here to plot the ROC curve corresponding to the aforementioned schemes. The general ROC curve of the proposed scheme has been plotted according to the average values of the data of the individual ROC curves shown in Fig. 1. Ten threshold values including the introduced optimal ones are used for drawing the ROC curve as shown in Fig. 2 with the magnified curve in Fig. 3. As Fig. 3 shows, the proposed scheme outperforms the other compared schemes in distinguishing different images (FAR is around 0). The TAR of the 2-phase scheme is relatively high, although it is shown that the TAR of the recently proposed scheme of [34] for a lower expected robustness is higher. It is worth mentioning that the scheme proposed in [34] is a keyless

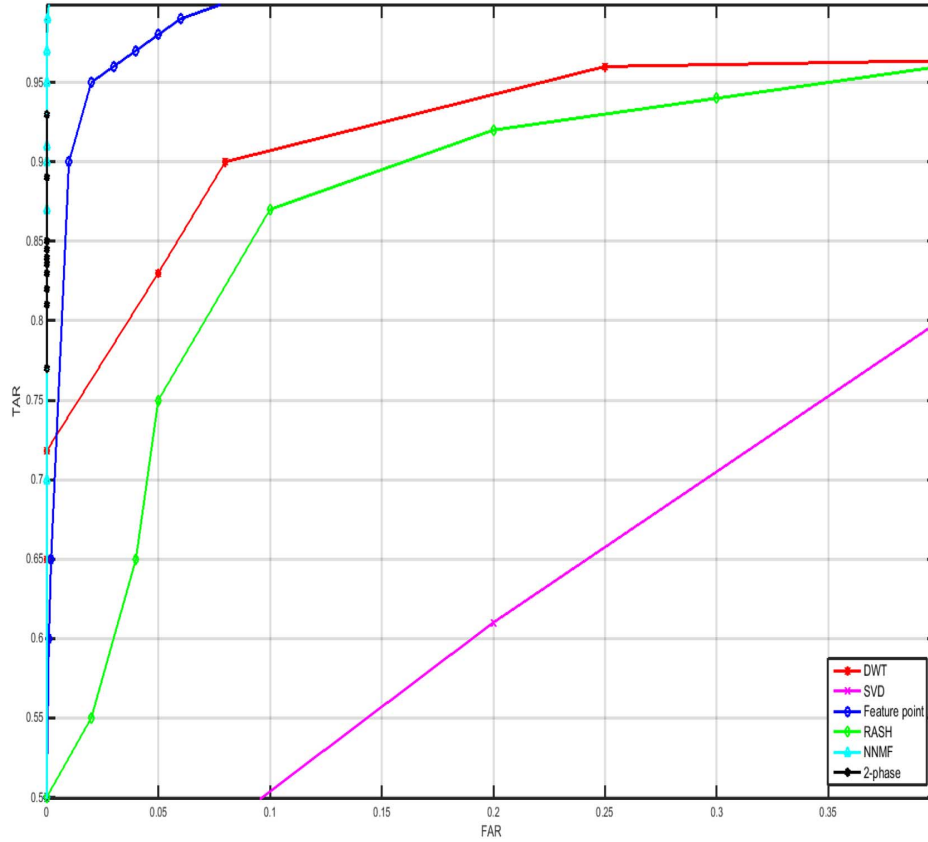


Fig. 3. Zoomed ROC curve of Fig. 2.

TABLE I
IMPLEMENTATION PARAMETERS OF THE TWO-PHASE SCHEME

Parameter	Value
T	DCT
Approximate correctness (P)	0.95
M, N	8
Number of blocks (t)	256
t_1	8
$ T_i^* $	8
Quantization factor	32

scheme and does not provide security. The robustness can be increased by adjusting the quantization factors at the cost of reduction of fragility and security. This issue will be addressed in the next section.

E. Detecting the Content Changes

To observe the sensitivity of the proposed scheme to changes in the image content, we have tested the proposed 2-phase image authentication scheme based on the implementation parameters presented in Table I against the most common types of image forgery attacks on the sample images shown in Figs. 4(a)–7(d). All forgery attacks contain semantic content modification, including object removal, object insertion, object modification, and exceeding the noise level. The reference and test images have been acquired using the above comparable hashing methods based on the optimal or recommended threshold and the distance function. The corresponding distance function, for which the optimal thresholds have been calculated, is the difference norm of the reference hash and the

received hash, except in the case of the NNMF-based method, in which a similarity measure is used [34]. The result is shown in the Table II. In another experiment the same image sample is subjected to a combination of attacks as shown in Fig. 8(a)–Fig. 8(d), yielding a similar result as shown in Table III. It may be seen from these results that DWT-based, SVD-based, and Radon-based schemes could not discriminate between the reference and test images. However, feature point based hash and NNMF-based hash could detect some noise exceeding attack and object modification cases as more global and local features of the image are captured via the recent schemes [34], [37].

The 2-phase approach is able to discriminate forged images from reference ones based on the the threshold factor $Tr = 60\%$. The reason for setting the threshold value to 60% comes from the fact that this lower bound guarantees the approximate correctness to be lower bounded by the preset value P . The details can be found in Section VI and [19]. The 2-phase hashing scheme is implemented based on the adjustable parameters given in Table I for which the corresponding ROC curves have been created.

VI. SECURITY AND PERFORMANCE ANALYSIS

In this section, security and performance studies of 2-phase image authentication scheme are given. The comprehensive security analysis of the corresponding building blocks is given in [19] based on computational security method for individual binary messages. To avoid repetition, we discuss the security



Fig. 4. Object moving. (a) Original image. (b) One moved object. (c) Two moved objects. (d) Three moved objects.



Fig. 5. Object deletion. (a) Original image. (b) One deleted object. (c) Two deleted objects. (d) Three deleted objects.

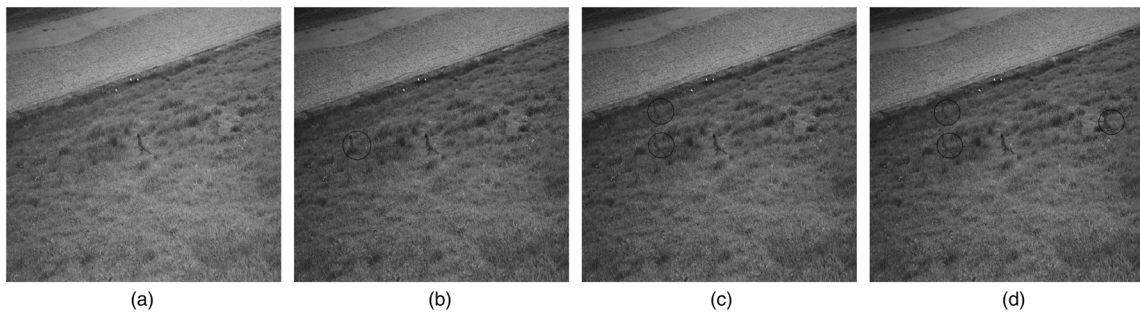


Fig. 6. Object addition. (a) Original image. (b) One added object. (c) Two added objects. (d) Three added objects.

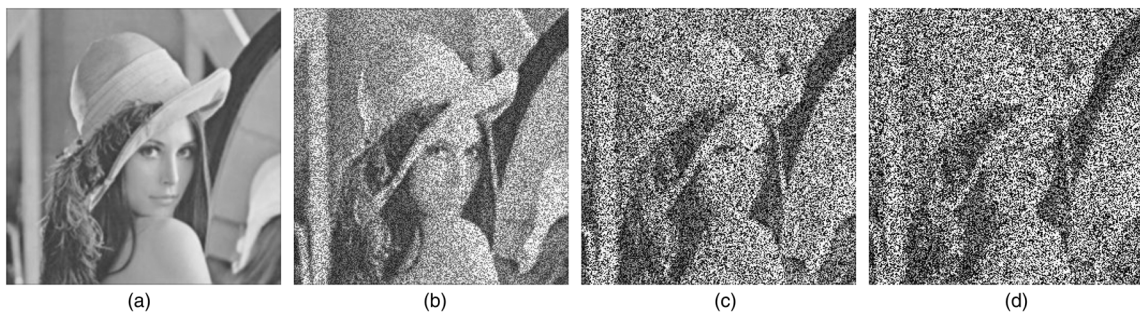


Fig. 7. Noise level. (a) Original image. (b) Exceeded noise level. (c) Exceeded noise level. (d) Exceeded noise level.

issues when an image feature is fed into the image authentication scheme. Another security aspect is evaluating key dependency of the tag generation algorithm. Strong statistical dependency of an image tag on the secret key is an important security feature and lack of such a dependency can allow compromising attacks. This issue is evaluated experimentally for 2-phase image authentication and the result is compared with other selected key-dependent schemes. The performance of

2-phase image authentication is evaluated analytically in terms of finding the threshold value such that the desired accuracy and security is achieved. This property makes the scheme more useful and promising.

A. Security Analysis

Security analysis of the proposed 2-phase image authentication scheme is based on the security aspects of the principal

TABLE II
FORGERY DETECTION COMPARISON FOR A SINGLE ATTACK

Hashing algorithm	Object modification (Figures 4a- 4d)	Object removal (Figures 5a- 5d)	Object addition (Figures 6a- 6d)	Exceeding noise level (Figures 7a- 7d)
DWT-based [35]	0.0057, 0.0061, 0.0061	0, 0.0065, 0.0071	0.0029, 0.0069, 0.0083	0.0183, 0.0519, 0.0634
SVD-based [31]	0.000076, 0.000081, 0.00009	0.00011, 0.000066, 0.00051	0.000035, 0.0017, 0.0021	0.00087, 0.0023, 0.0028
Feature point based [36]	0.0172, 0.0477, 0.0487	0.0076, 0.0063, 0.0074	0.0028, 0.0166, 0.0279	0.4009, 0.9272, 0.9911
Radon-based [38]	0.00065, 0.00066, 0.00065	0.00025, 0.00024, 0.00045	0, 0.0015, 0.0016	0.0083, 0.044, 0.0588
NNMF-based [34]	0.9056 , 0.9961, 0.9946	0.9677, 0.9826, 0.9350	0.9864, 0.9987, 0.9978	0.7803, 0.2688, 0.2688

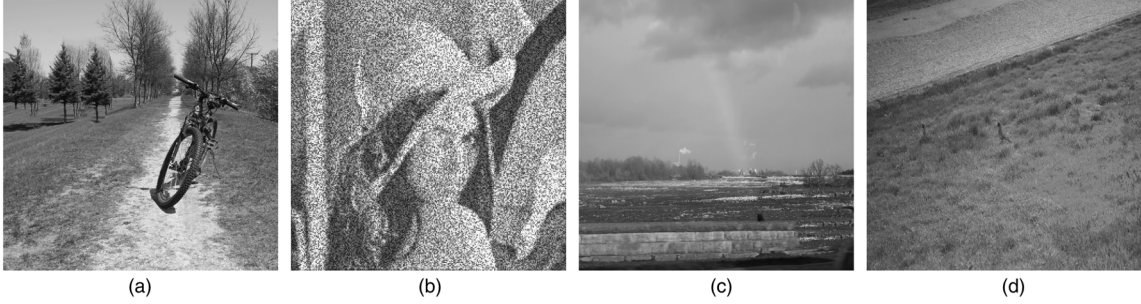


Fig. 8. Combined modifications. (a) Deletion and moving. (b) Noising and moving. (c) Deletion and moving. (d) Adding and moving.

TABLE III
FORGERY DETECTION COMPARISON FOR COMBINED ATTACKS

Hashing algorithm	Figure 5a and Figure 8a	Figure 7a and Figure 8b	Figure 4a and Figure 8c	Figure 6a and Figure 8d
DWT-based [35]	0.0176	0.017	0.0096	0.10136
SVD-based [31]	0.0036	0.001	0.0011	0.000271
Feature point based [36]	0.0337	0.3359	0.3937	0.0429
Radon-based [38]	0.0046	0.0085	0.0046	0.003
NNMF-based [34]	0.9989	0.7	0.4776	0.9142

components $AMAC_1$ and $AMAC_2$ for providing hard and soft authentication respectively. The image features containing the most compact and essential information about the image are protected by $AMAC_1$ while the feature part which contains less information is supported by $AMAC_2$. To consider the performance of the proposed scheme, the following random variables are defined:

$$F_i = \begin{cases} 1 & \text{if } T_i \neq T'_i \\ 0 & \text{if } T_i = T'_i \end{cases}, \quad F = \sum_{i=1}^{t_2} F_i.$$

Using the classical occupancy problem [39], the probability P_F that one non-matched block appears in the second subtag is calculated as follows:

$$P_F = Pr(F_i = 1) = 1 - \left(\left(1 - \left(\frac{\binom{p^2 + \delta - 2}{\delta}}{\binom{p^2 + \delta - 1}{\delta}} \right)^{t_1} \right) 2^{-|T_i|} + \left(\frac{\binom{p^2 + \delta - 2}{\delta}}{\binom{p^2 + \delta - 1}{\delta}} \right)^{t_1} \right). \quad (3)$$

To obtain a desirable level of security and accuracy, it is possible to find a lower bound for the number of required hash-blocks by initializing the other adjustable parameters. Therefore, the

probability of the appearance of a smaller-than-threshold value of non-matched blocks ($\lceil Tr \cdot t_2 \rceil$) must be calculated. The corresponding random variable F follows a binomial distribution with parameters t_2 and P_F . After applying the Chernoff inequality [19], [40] ($Pr(F < Tr \cdot t_2) \leq e^{-2t_2(P_F - Tr)^2}$) and robustness requirement ($Pr(F < Tr \cdot t_2) \leq 1 - P$), a lower bound for the threshold ratio can be extracted as follows:

$$Tr \geq P_F + \sqrt{\frac{1}{2t_2} \ln \frac{1}{1 - P}}. \quad (4)$$

A potential threat against proposed 2-phase scheme is the impact of an image tampering attack resulting in a forged image. The verification could be successful on a valid forged image which is not authentic if the tampered pixels are not reflected by the protected components in the first phase of verification. Protecting enough feature elements of the transformation T from each image block (or ring) makes this attack scenario computationally impossible. This is not proven for the proposed scheme and a comprehensive analysis regarding the impact of image tampering is required. However, this threat is not applicable in the second verification phase as the image thumbnail is protected by $AMAC_2$.

Another attack scenario is to modify the image content such that the extracted elements from the transformation T can be restored via error-correcting code in the first phase. This attack is the most powerful attack on the 2-phase scheme as it can be mounted without any complexity and only requires offline computations. To prevent this attack, the tag generation part corresponding to $AMAC_1$ is modified: some least significant

TABLE IV
KEY DEPENDENCY COMPARISON

Hashing algorithm	Distance range	Mean	Variance
DWT-based [35]	[59%, 61%]	60.38%	644.03×10^{-5}
SVD-based [31]	[5.5%, 8.4%]	6.62%	4.1412×10^{-5}
2-phase scheme	[49.3%, 51.4%]	50.46%	8.5121×10^{-6}

bits (LSBs) of the selected elements of T for the first phase are considered only by $AMAC_1$ for correction via error-correcting code. However, the MAC of all selected elements is computed in $AMAC_1$. For example, in the DCT-RS-based implementation, the first 4 LSBs of each DC can be used for encoding and encryption while the MAC of all DCs are computed to generate the first part of the tag. This improvement allows the less fragile part of the elements to be corrected and avoids intentional changes which can be influenced by content modifications.

B. Dependency on the Secret Key

One of the most important security concerns in image hashing mechanisms is the sensitivity of the tag generation algorithm to key changes. From a cryptographic viewpoint, any change in the secret key should be detectable by the corresponding change of approximately 50% of the output bits. Any observed bias can significantly reduce the complexity of the key recovery attack as well as the lifetime of the secret key. Using cryptographic MACs in both phases guarantees such a desirable security performance. Here, we provide experimental results which compare the key dependency characteristics between the proposed 2-phase hash generation algorithm and the approaches described above. In each experiment, the secret key is changed by one bit and the difference between the outputs is calculated. The first method [35] uses secret keys in two steps including random image tiling and randomized rounding. The second method [31] uses three secret keys to generate the final hash value. The feature point based approach [36] and Radon-based method [38] (deterministic version only) do not use a secret key in the tag generation algorithm, so that the experiment is not applicable in this case. The final approach, which is based on the ring partitioning and NNMF, uses initial values to find the decomposition factors [34]. So, the latter image hashing scheme is also considered a keyless approach. The experimental results are shown in the Table IV. According to the results, the proposed 2-phase image authentication method shows almost ideal behavior regarding the key dependency issue. Among the compared schemes, the SVD-based method [31] does not show a desired sensitivity to key changes. This point indicates that the effective key length is less than the actual key length used in the algorithm. The other scheme also does not show almost ideal sensitivity against key changes. Using the data of input image in the key insertion step of the tag generation algorithm is a solution to increase the effective key length significantly.

VII. CONCLUSION

A generic robust and secure 2-phase scheme for image authentication has been presented. The verification algorithm verifies the authenticity of the received image in two phases based on the combination of soft and hard authentication techniques.

The user may adjust the parameters to achieve the desired robustness and security. The proposed scheme was shown to be tolerant to common content-preserving modifications on images which may be introduced during transmission or which may be the result of image processing operations. Robustness against geometric distortions, JPEG compression, filtering operations, and additive noise was demonstrated via ROC curves. A weakness against rotation was discovered and a solution based on a recently published scheme was proposed.

A lower bound for the threshold value was calculated to achieve a desired accuracy and security. Potential security flaws of the scheme were demonstrated and some solutions proposed. The proposed 2-phase image authentication almost outperforms the compared schemes in the context of forgery attack detection as shown in Tables II and III. However, it is unable to classify and identify the type of attack (e.g., [1]) or to localize the attacked part of the image. Future work will be focused on improvement of the scheme towards the tamper/forgery classification and accurate forgery area localization of images.

ACKNOWLEDGMENT

The authors would like to thank M. Flanagan for his helpful language editing assistance. The authors are also grateful for the anonymous reviewers and the associate editor for their valuable comments and suggestions.

REFERENCES

- [1] Y. Zhao, S. Wang, X. Zhang, and H. Yao, "Robust hashing for image authentication using zernike moments and local features," *IEEE Trans. Inf. Forensics Security*, vol. 8, no. 1, pp. 55–63, Jan. 2013.
- [2] D. Tonien, R. Safavi-Naini, P. Nickolas, and Y. Desmedt, "Unconditional secure approximate message authentication," in *Proc. Int. Workshop Coding Cryptology*, 2009, vol. 5557, pp. 233–247.
- [3] B. B. Zhu, M. D. Swanson, and A. H. Tewfik, "When seeing is't believing," *IEEE Signal Process. Mag.*, vol. 21, no. 2, pp. 40–49, Mar. 2004.
- [4] A. Haouzia and R. Noumeir, "Methods for image authentication: A survey," *Multimedia Tools Appl.*, vol. 39, pp. 1–46, 2008.
- [5] J. Shin and C. Ruland, "A survey of image hashing technique for data authentication in WMSNs," in *Proc. IEEE Int. Conf. Wireless Mobile Comput., Netw. Commun.*, Oct. 2013, pp. 253–258.
- [6] A. Menezes, P. Van Oorschot, and S. Vanstone, *Handbook of Applied Cryptography*. Boca Raton, FL, USA: CRC Press, 1996.
- [7] J. Fridrich and M. Goljan, "Protection of digital images using self embedding," in *Proc. Symp. Content Security Data Hiding Digital Media*, 1999.
- [8] J. Fridrich, "Methods for tamper detection in digital images," in *Proc. Multimedia Security Workshop ACM Multimedia*, Orlando, FL, USA, 1999, pp. 29–33.
- [9] S. C. Byun, I. L. Lee, and T. H. Shin, "A public-key based watermarking for color image authentication," in *Proc. IEEE Int. Conf. Multimedia Expo.*, Aug. 2002, vol. 1, pp. 593–600.
- [10] L. Xie, G. R. Arce, and R. F. Graveman, "Approximate image message authentication codes," *IEEE Trans. Multimedia*, vol. 3, no. 2, pp. 242–252, Jun. 2001.
- [11] L. Xie and A. Lewis, "Methods for soft image/video authentication," in *Proc. Annu. Fedlab Symp. ATIRP*, 2000.
- [12] Z.-M. Lu, D.-G. Xu, and S.-H. Sun, "Multipurpose image watermarking algorithm based on multistage vector quantization," *IEEE Trans. Image Process.*, vol. 14, no. 6, pp. 822–832, Jun. 2005.
- [13] N. E. D. Sherif and M. Mansour, "Fragile and semi-fragile image authentication based on image self similarity image processing," in *Proc. Int. Conf. Image Process.*, 2002, vol. 2, pp. 897–900.
- [14] F. Jing, M. Li, H. J. Zhang, and B. Zhang, "An efficient and effective region-based image retrieval framework," *IEEE Trans. Image Process.*, vol. 13, no. 5, pp. 699–709, May 2004.
- [15] M. P. Queluz, "Toward robust, content based techniques for image authentication," in *Proc. Workshop Multimedia Signal Process.*, 1998, pp. 297–302.

- [16] C. Kailasanathan, R. Safavi-Naini, and P. Ogunbona, "Image authentication surviving acceptable modifications," in *Proc. IEEE-EURASIP, Workshop Nonlinear Signal Image Process.*, 2001.
- [17] I. C. Chang, B. W. Hsu, and C. S. Lai, "A DCT quantization-based image authentication system for digital forensics," in *Proc. Int. Workshop Systematic Approaches Digit. Forensic Eng.*, 2005, pp. 223–235.
- [18] H. T. Chang, C.-C. H., C.-H. Y., and D.-F. S., "Image authentication with tampering localization based on watermark embedding in wavelet domain," *Opt. Eng.*, vol. 48, no. 5, pp. 1–10, 2009.
- [19] G. D. Crescenzo, R. F. Graveman, and G. R. Arce, "Approximate message authentication and biometric entity authentication," in *Proc. Financial Cryptography*, 2005, vol. 3570, pp. 240–254.
- [20] R. F. Graveman and K. Fu, "Approximate message authentication codes," in *Proc. Annu. Symp. Adv. Telecommun. Inf. Distrib. Res. Program*, 1999.
- [21] R. Ge, G. R. Arce, and G. D. Crescenzo, "Approximate message authentication codes for N-ary alphabets," *IEEE Trans. Inf. Forensics Security*, vol. 1, no. 1, pp. 56–67, Mar. 2006.
- [22] D. Tonien, R. Safavi-Naini, and P. Nickolas, "Breaking and repairing an approximate message authentication scheme," *Discrete Math., Algorithms Appl.*, vol. 3, no. 3, pp. 393–412, 2011.
- [23] R. Safavi-Naini and D. Tonien, "Fuzzy universal hashing and approximate authentication," *Discrete Math., Algorithms Appl.*, vol. 3, no. 4, pp. 587–607, 2011.
- [24] R. Safavi-Naini and D. Tonien, "Fuzzy universal hashing and approximate authentication," *Cryptography ePrint Archive* pp. 1–19, Aug. 2005 [Online]. Available: <http://eprint.iacr.org/2005/256>
- [25] S. A. H. A. E. Tabatabaei, O. Ur-Rehman, and N. Zivic, "AACI: A mechanism for authentication and correction of images," in *Proc. IEEE Int. Conf. Commun. Workshops*, Jun. 2013, pp. 727–732.
- [26] G. Zhu, J. Huang, S. Kwong, and J. Yang, "A study on the randomness measure of image hashing," *IEEE Trans. Inf. Forensics Security*, vol. 4, no. 4, pt. 2, pp. 928–932, Dec. 2009.
- [27] O. Koval, S. Voloshnovskiy, F. Beekhof, and T. Pun, "Security analysis of robust perceptual hashing," in *Proc. SPIE 6819, Security, Forensics, Steganography, Watermarking Multimedia Contents X*, 2008, p. 681906.
- [28] S. A. H. A. E. Tabatabaei, O. Ur-Rehman, and N. Zivic, "A DWT-based image authentication and correction mechanism," in *Proc. Int. Symp. ELMAR*, 2013, 2013, pp. 79–82.
- [29] M. Naor and M. Yung, "Universal one-way hash functions and their cryptographic applications," in *Proc. Annu. ACM Symp. Theory Comput.*, 1989, pp. 33–43.
- [30] Z. Tang, S. Wang, X. Zhang, W. Wei, and S. Su, "Robust image hashing for tamper detection using non-negative matrix factorization," *J. Ubiquitous Convergence Technol.*, vol. 2, no. 1, pp. 18–26, 2008.
- [31] S. S. Kozat, R. Venkatesan, and K. Mihcak, "Robust perceptual image hashing via matrix invariants," in *Proc. Int. Conf. Image Process.*, 2004, pp. 3443–3447.
- [32] I. S. Reed and G. Solomon, "Polynomial codes over certain finite fields," *SIAM J. Appl. Math.*, vol. 8, pp. 300–304, 1960.
- [33] A. Swaminathan, Y. Mao, and M. Wu, "Robust and secure image hashing," *IEEE Trans. Inf. Forensics Security*, vol. 1, no. 2, pp. 215–230, Jun. 2006.
- [34] Z. Tang, X. Zhang, and S. Zhang, "Robust perceptual image hashing based on ring partition and NMF," *IEEE Trans. Knowl. Data Eng.*, vol. 26, no. 3, pp. 711–724, Mar. 2014.
- [35] R. Venkatesan, S.-M. Koon, M. H. Jakubowski, and P. Moulin, "Robust image hashing," in *Proc. IEEE Int. Conf. Image Process.*, Sep. 2000, vol. 3, pp. 664–666.
- [36] V. Monga and B. L. Evans, "Perceptual image hashing via feature points: Performance evaluation and tradeoffs," *IEEE Trans. Image Process.*, vol. 15, no. 11, pp. 3452–3465, Nov. 2006.
- [37] K. Mihcak and R. Venkatesan, "New iterative geometric techniques for robust image hashing," in *Proc. ACM Work. Security Privacy Digit. Rights Manag.*, 2001, pp. 13–21.
- [38] F. Lefebvre, B. Macq, and J. Legat, "Rash: Radon soft hash algorithm," in *Proc. Eur. Signal Process. Conf.*, 2002, pp. 1–4.
- [39] N. Feller, *An Introduction to Probability Theory and Its Applications*, 3rd ed. Hoboken, NJ, USA: Wiley, 1968.
- [40] M. Mitzenmacher and U. Upfal, *Probability and Computing - Randomized Algorithms and Probabilistic Analysis*. Cambridge, U.K.: Cambridge Univ. Press, 2005.



Seyed Amir Hossein Tabatabaei received the B.Sc. degree in applied mathematics from the University of Tehran, Tehran, Iran, and the M.Sc. degree in applied mathematics from the University of Tarbiat Modares, Tehran, Iran.

From 2001 to 2009, he was a Lecturer and Researcher with various universities and R&D departments in electrical industries. Since December 2010, he has been a Research Assistant and Ph.D. candidate with the Chair for Data Communications Systems, University of Siegen, Siegen, Germany.



Obaid Ur-Rehman received the M.Sc. degree in computer engineering from the University of Engineering and Technology, Taxila, Pakistan, in 2004, and the Dr.-Ing degree from the University of Siegen, Siegen, Germany, in 2012.

He is currently a Post Doctoral Fellow with the Chair for Data Communications, University of Siegen. He has authored or coauthored more than 25 scientific research publications in international conferences and journals. His current research interests include error-correcting codes, communications, and

network security.

Dr. Ur-Rehman serves as the reviewer of various international journals.



Natasa Zivic received the Dipl.-Ing. degree and the Magister degree in electrical engineering from the University of Belgrade, Belgrade, Serbia, in 1999 and 2002, respectively, and the Dr.-Ing. degree in electrical and computer engineering from the University of Siegen, Siegen, Germany, in 2007.

She is currently an Assistant Professor and Private Docent with the University of Siegen. She authored or coauthored about 100 journal and conference papers, two monographs, and three patents (two in Germany and one in the U.S.). Her current research inter-

ests include the connection between cryptography and standard communication techniques like channel coding, and applying of cryptography in communications, especially in noisy environments.

Dr. Zivic is a member of ISO SC 27 Security Techniques and acted as a project editor of security standards. She served as a TPC member and an organizer of several IEEE conferences and workshops, and has also been a technical reviewer of several journals.



Christoph Ruland received the Diploma degree and Dr. degree in mathematics from the University of Bonn, Bonn, Germany, in 1974 and 1976, respectively.

He worked for six years in the industry developing communication protocols and managing network projects and products. In 1982, he became a Professor of Telecommunication with the Aachen University of Applied Sciences, Aachen, Germany. In 1988, he founded KryptoKom, Aachen, Germany. In 1992, he became a Full Professor of Data

Communications Systems with the Department for Electrical Engineering and Computer Science, University of Siegen, Siegen, Germany. He has authored or coauthored more than 140 research papers and two books. He holds several patents in data communications, positioning, and eMobility. His research interests include the integration of cryptography and security into communications systems.

Dr. Ruland is member of ISO/IEC SC 27 Security Techniques.